Relations and Functions

Relations and Functions

You'll Learn To:

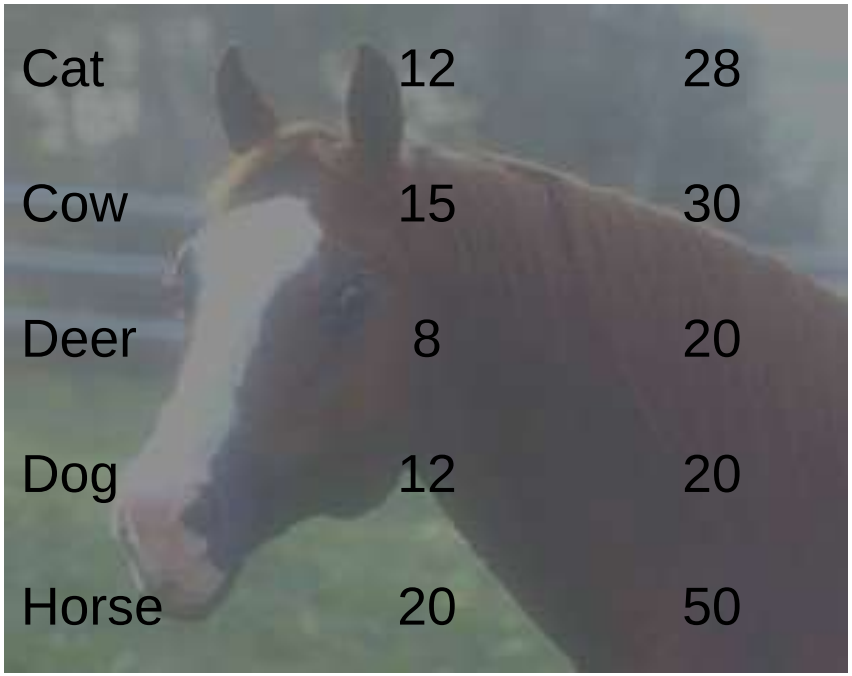
- Analyze and graph relations.
- Find functional values.

Vocabulary

1) ordered pair
2) Cartesian Coordinate
3) plane
4) quadrant
5) relation
6) domain
7) range
8) function
9) mapping
10) one-to-one function
11) vertical line test
12) independent variable
13) dependent variable
14) functional notation

Relations and Functions

This table shows the average lifetime and maximum lifetime for some animals.



Cat	12	28
Cow	15	30
Deer	8	20
Dog	12	20
Horse	20	50

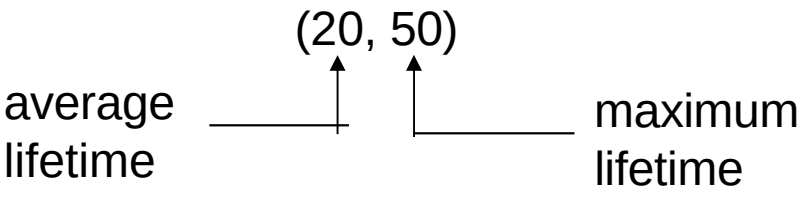
The data can also be represented as **ordered pairs**.

The ordered pairs for the data are:

(12, 28), (15, 30), (8, 20),

(12, 20), and (20, 50)

The first number in each ordered pair is the average lifetime, and the second number is the maximum lifetime.

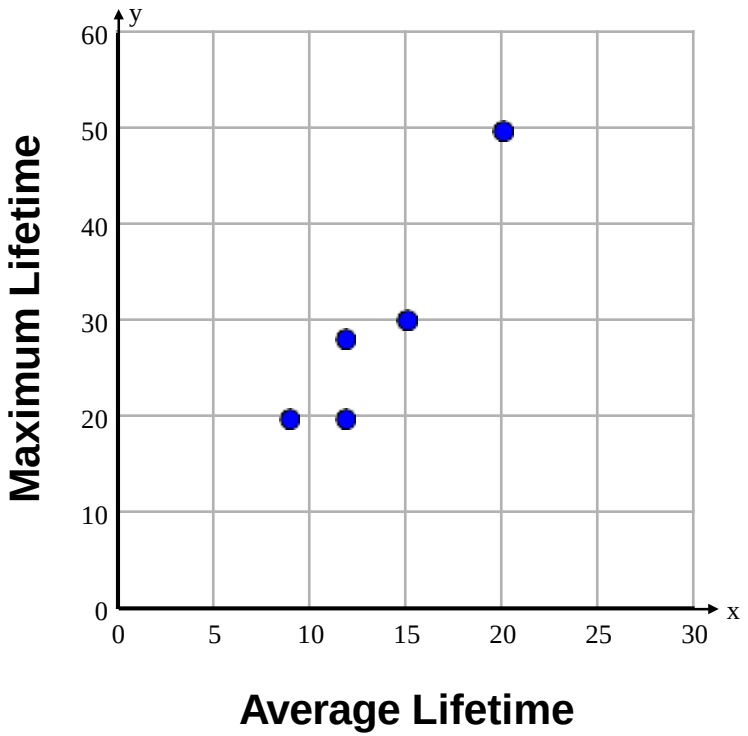


Relations and Functions

You can graph the ordered pairs below on a ***coordinate system*** with two axes.

(12, 28), (15, 30), (8, 20),

(12, 20), and (20, 50)



Remember, each point in the coordinate plane can be named by **exactly one ordered pair** and that every ordered pair **names exactly one point** in the coordinate plane.

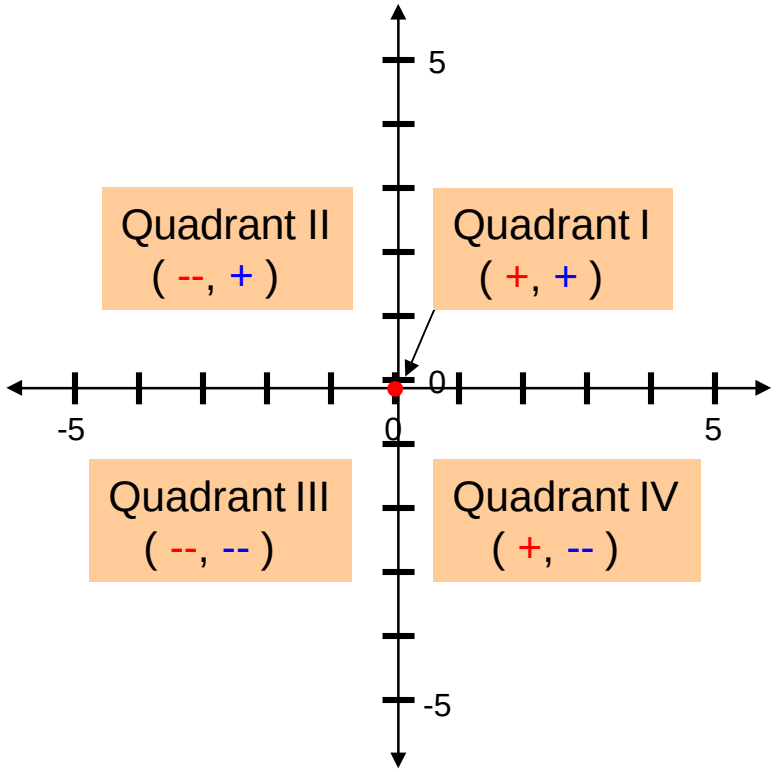
The graph of this data (animal lifetimes) lies in only one part of the **Cartesian** coordinate plane – the part with all positive numbers.

Relations and Functions

The **Cartesian coordinate system** is composed of the x-axis (horizontal),

and the y-axis (vertical), which meet at the origin (0, 0) and divide the plane into four quadrants.

You can tell which quadrant a point is in by looking at the sign of each coordinate of the point.



The points on the two axes do not lie in any quadrant.

In general, any ordered pair in the coordinate plane can be written in the form (x, y).

What is a RELATION?

A **relation** is a set of ordered pairs.

The **domain** of a relation is the set of all first coordinates (*x-coordinates*) from the ordered pairs.

The **range** of a relation is the set of all second coordinates (*y-coordinates*) from the ordered pairs.

The ***graph*** of a relation is the set of points in the coordinate plane corresponding to the ordered pairs in the relation.

Given the relation:

{(2, -6), (1, 4), (2, 4), (0,0), (1, -6), (3, 0)}

State the domain:

D: {0,1, 2, 3}

State the range:

R: {-6, 0, 4}

Note: { } are the symbol for "set".

When writing the domain and range, do not repeat values.

State the domain and range of the following relation.

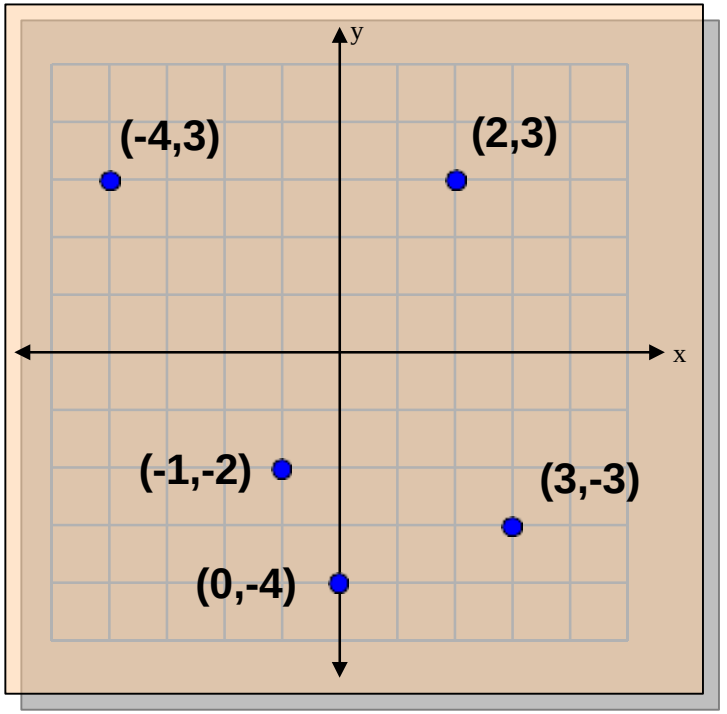
{(2, –3), (4, 6), (3, –1), (6, 6), (2, 3)}

domain: {2, 3, 4, 6}

range: {–3, –1, 3, 6}

Relations and Functions

State the domain and range of the relation shown in the graph.



The **relation** is:

{ (-4, 3), (-1, -2), (0, -4),
(2, 3), (3, -3) }

The **domain** is:

{ -4, -1, 0, 2, 3 }

The **range** is:

{ -4, -3, -2, 3 }

Relations and Functions

- Relations can be written in several ways: ordered pairs, table, graph, or mapping.

- We have already seen relations represented as ordered pairs.

Table

{(3, 4), (7, 2),
(0, -1), (-2, 2),
(-5, 0), (3, 3)}

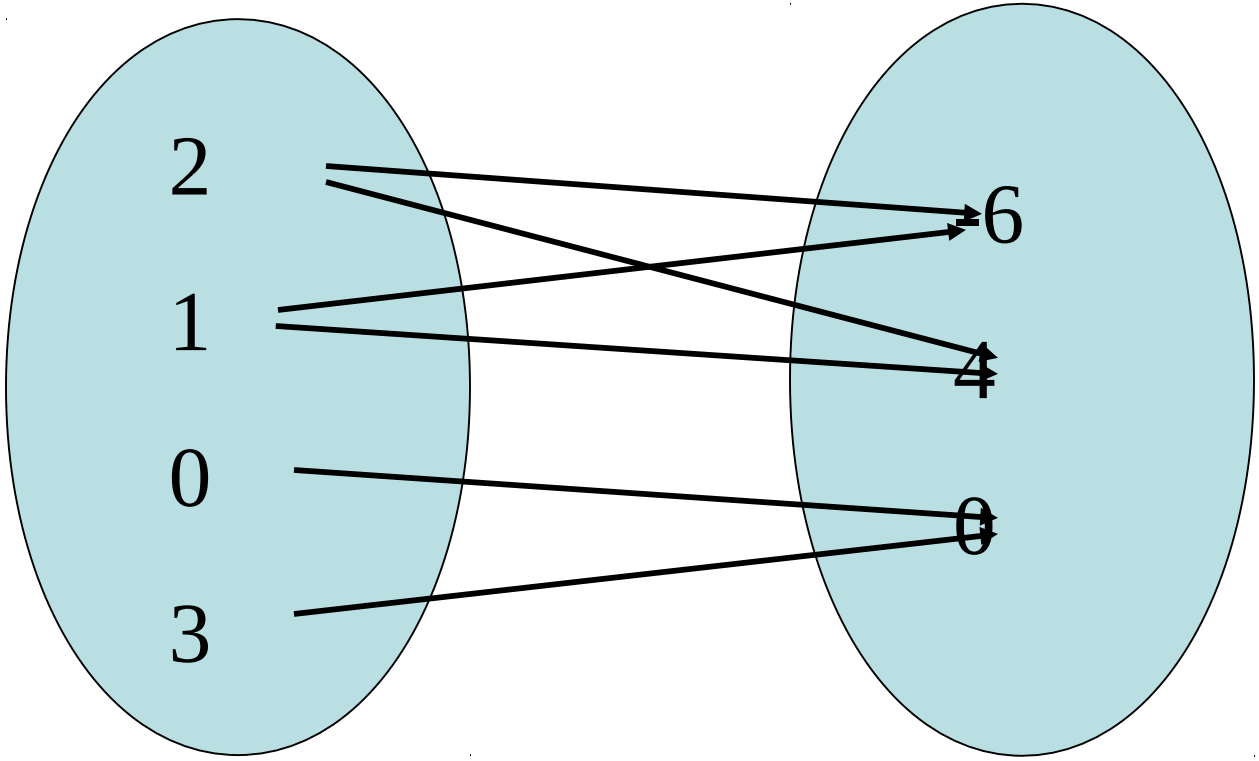
x	y
3	4
7	2
0	-1
-2	2
-5	0
3	3

Mapping

- Create two ovals with the domain on the left and the range on the right.
- Elements are not repeated.
- Connect elements of the domain with the corresponding elements in the range by drawing an arrow.

Mapping

{(2, -6), (1, 4), (2, 4), (0, 0), (1, -6), (3, 0)}



In summary:

RELATIONS

- A **relation** is a set of pairs of input and output values.
  - There are four ways to represent relations

take note

Key Concept Four Ways to Represent Relations

Ordered Pairs

(input, output)

(x, y)
$(-3, 4)$
$(3, -1)$
$(4, -1)$
$(4, 3)$

Mapping Diagram

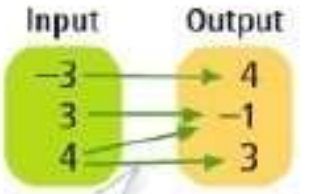
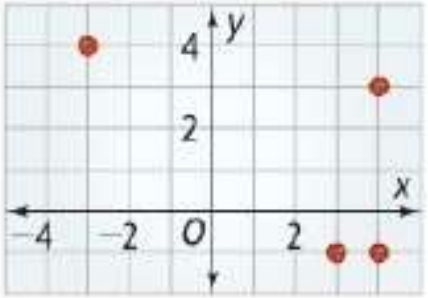


Table of Values

x Input	y Output
−3	4
3	−1
4	−1
4	3

Graph



FUNCTIONS

Objective: To recognize whether a relation is a function or not.

What is a FUNCTION?

Relations and Functions

A **function** is a special type of relation in which each element of the domain is paired with **exactly one** element in the range.

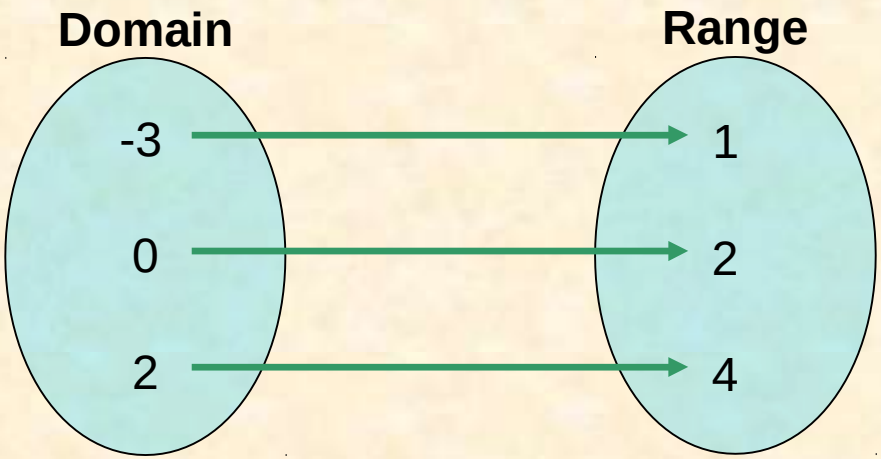
Furthermore, a set of ordered pairs is a function if no two ordered pairs have equal abscissas.

A **mapping** shows how each member of the domain is paired with each member in the range.

Functions

$$\{(-3,1),(0,2),(2,4)\}$$

ONE-TO-ONE CORRESPONDENCE



function

Relations and Functions

A **function** is a special type of relation in which each element of the domain is paired with _**exactly one** element in the range.

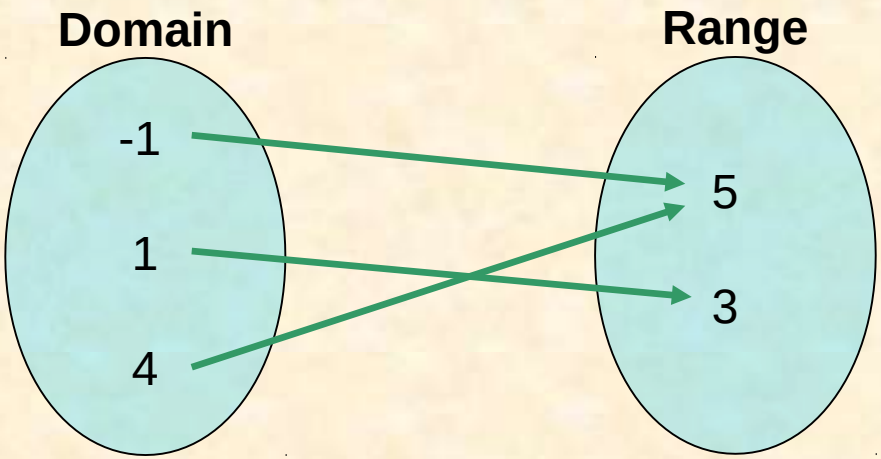
Furthermore, a set of ordered pairs is a function if no two ordered pairs have equal abscissas.

A **mapping** shows how each member of the domain is paired with each member in the range.

Functions

$$\{(-1,5), (1,3),(4,5)\}$$

MANY-TO-ONE CORRESPONDENCE



function, not one-to-one

Relations and Functions

A **function** is a special type of relation in which each element of the domain is paired with **exactly one** element in the range.

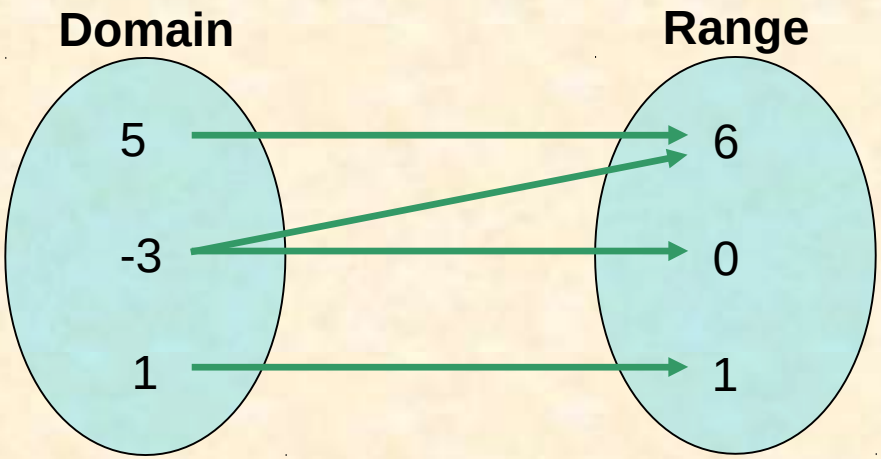
Furthermore, a set of ordered pairs is a function if no two ordered pairs have equal abscissas.

A **mapping** shows how each member of the domain is paired with each member in the range.

Functions

$$\{(5,6),(-3,0),(1,1),(-3,6)\}$$

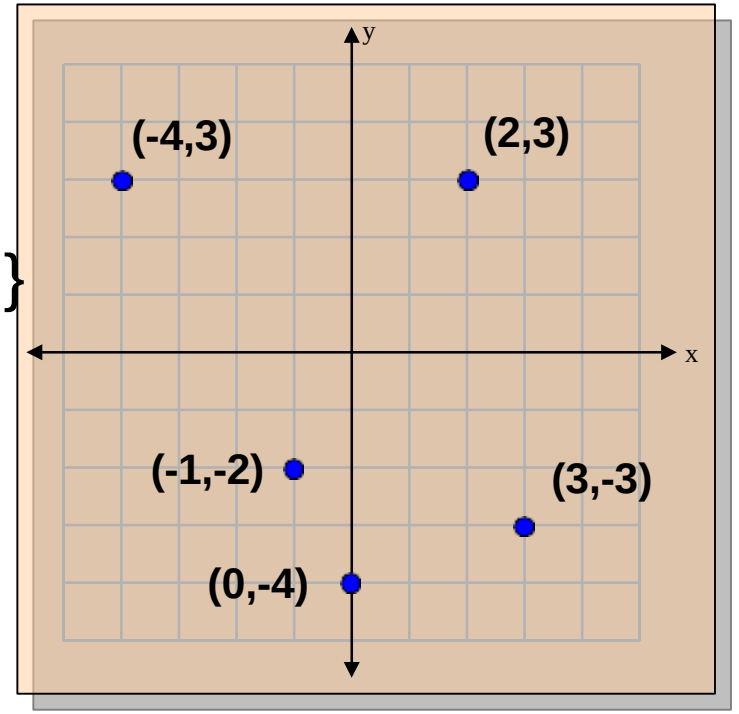
ONE-TO-MANY CORRESPONDENCE



not a function

Relations and Functions

State the domain and range of the relation shown in the graph. Is the relation a function?



The **relation** is:

{ (-4, 3), (-1, 2), (0, -4), (2, 3), (3, -3) }

The **domain** is:

{ -4, -1, 0, 2, 3 }

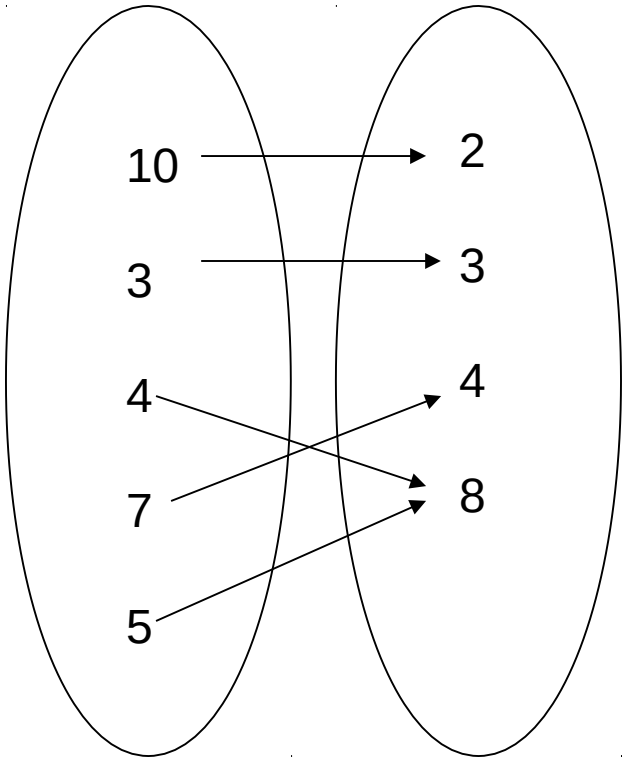
The **range** is:

{ -4, -3, -2, 3 }

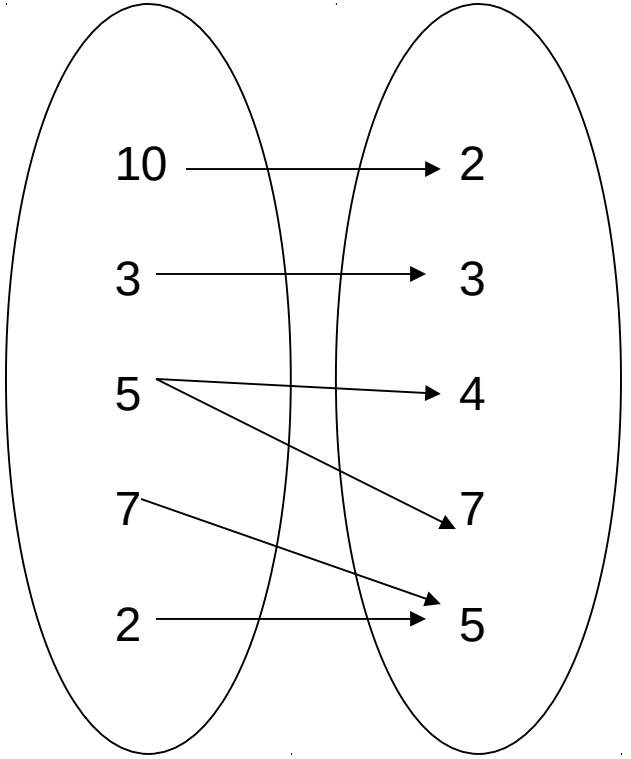
Each member of the domain is paired with Exactly one member of the range, so this relation is a function.

Function	Not a Function
(4,12)	(4,12)
(5,15)	(4,15)
(6,18)	(5,18)
(7,21)	(5,21)
(8,24)	(6,24)

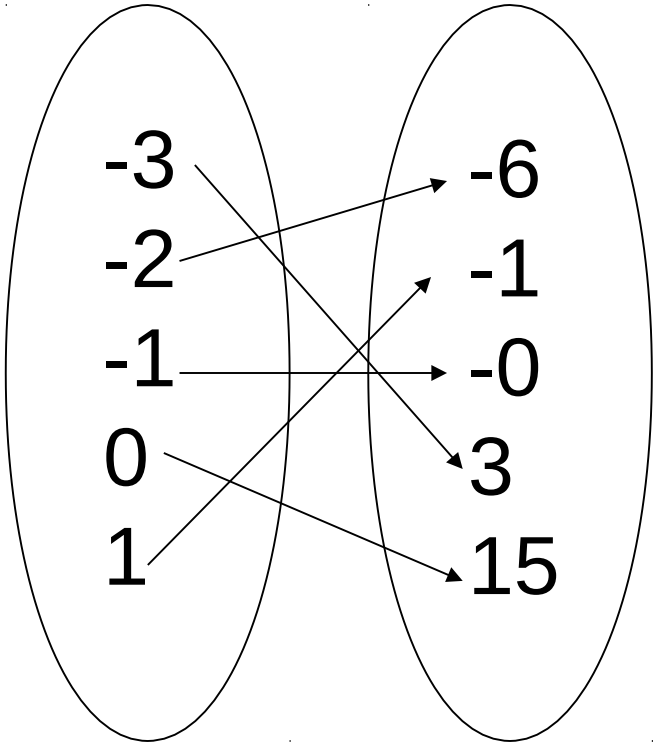
Function



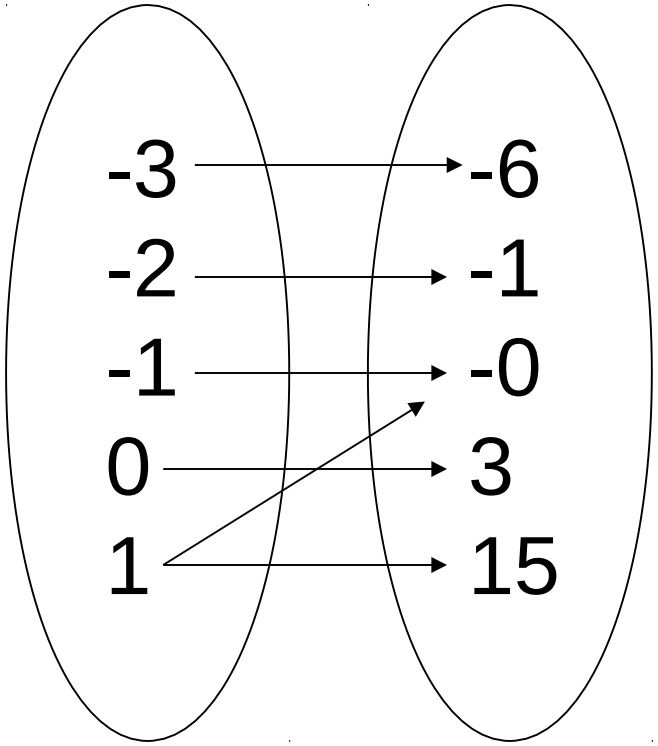
Not a Function



Function



Not a Function



Function

X	Y
1	2
2	4
3	6
4	8
5	10
6	12

Not a Function

X	Y
1	2
2	4
1	5
3	8
4	4
5	10

Function

X	-3	0	3	8	-10
Y	6	8	20	4	8

Not a Function

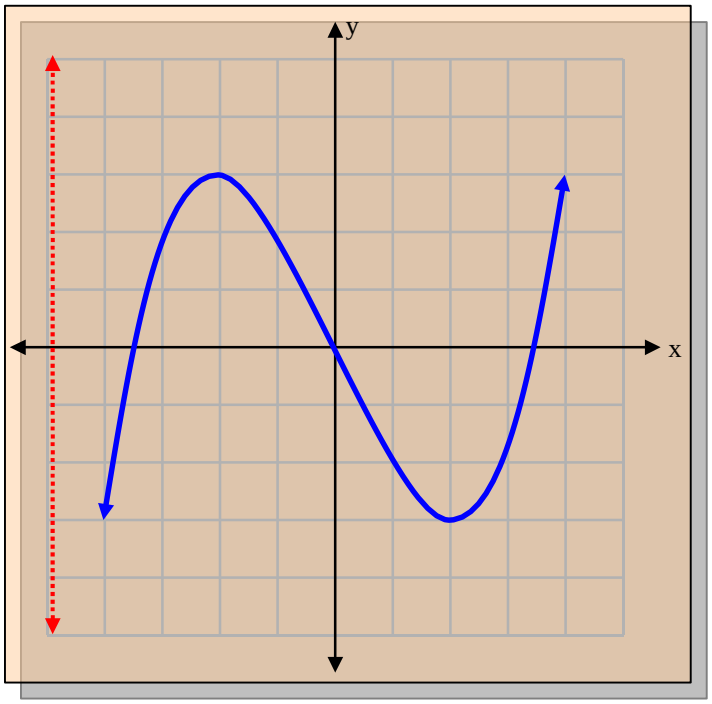
X	-2	0	-2	7	-8
Y	6	8	20	4	8

Relations and Functions

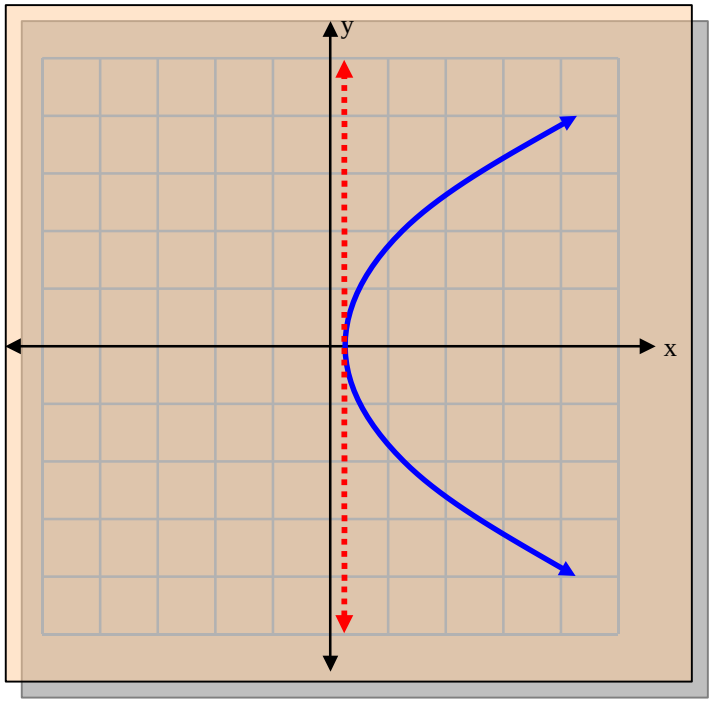
You can use the **vertical line test** to determine whether a relation is a function.

Vertical Line Test

If no vertical line intersects a graph in more than one point, the graph represents a function.



If some vertical line intercepts a graph in two or more points, the graph does not represent a function.

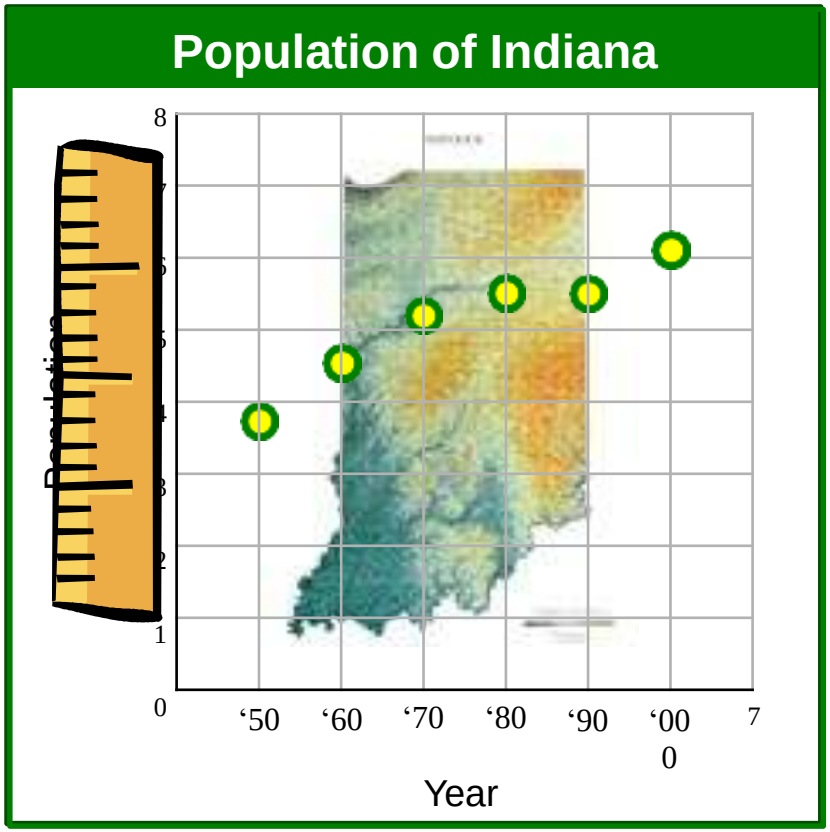


Relations and Functions

The table shows the population of Indiana over the last several decades.

Year	Population (millions)
1950	3.9
1960	4.7
1970	5.2
1980	5.5
1990	5.5
2000	6.1

We can graph this data to determine if it represents a function.

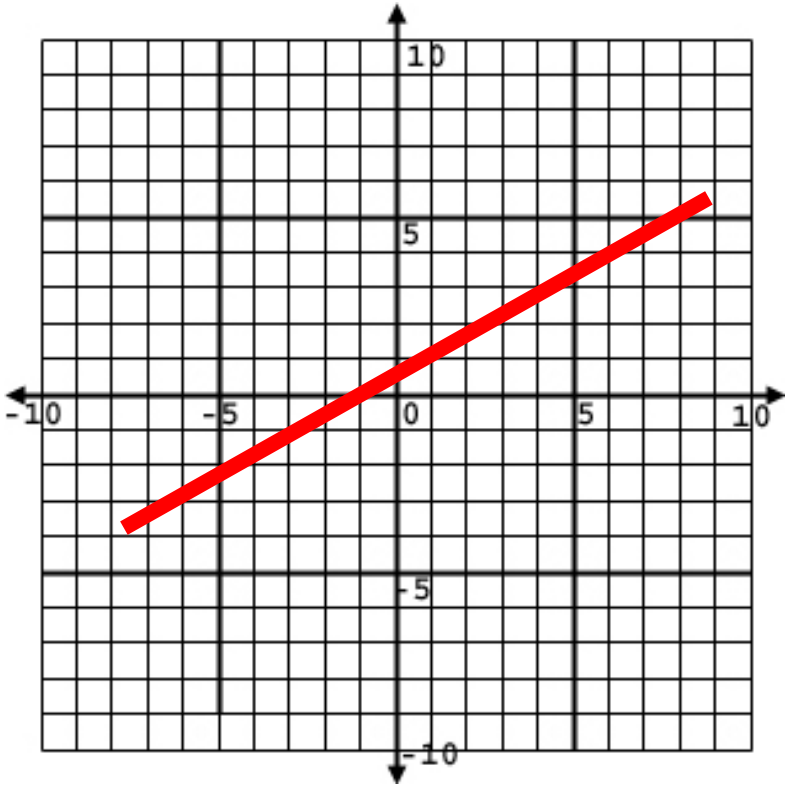


Use the vertical line test.

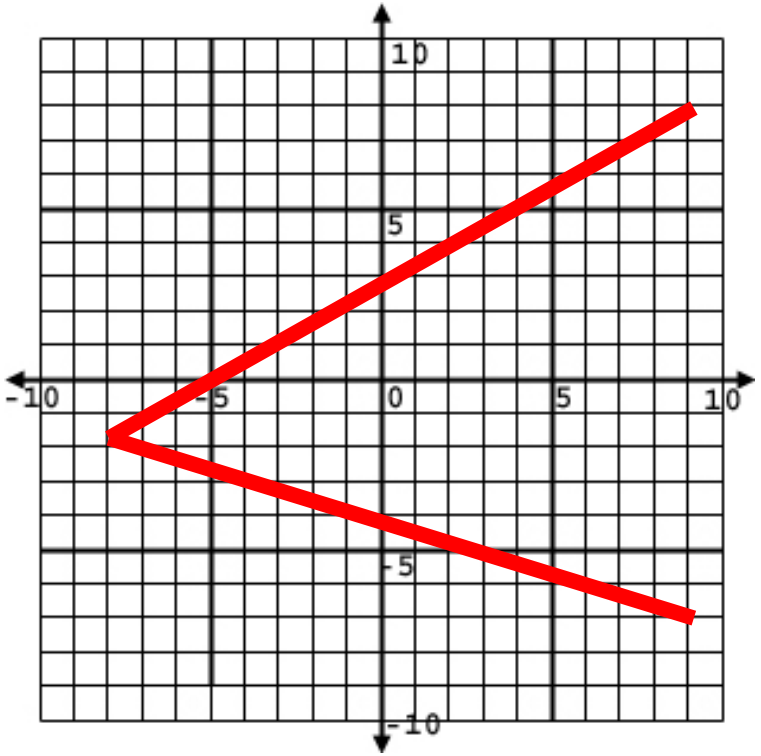
Notice that no vertical line can be drawn that contains more than one of the data points.

Therefore, this relation is a function!

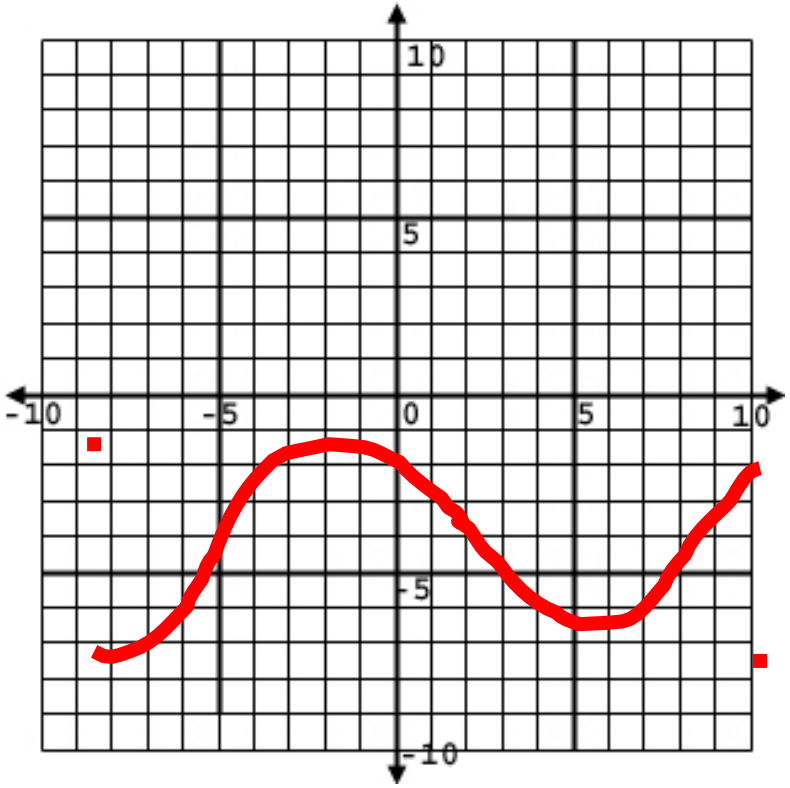
Function



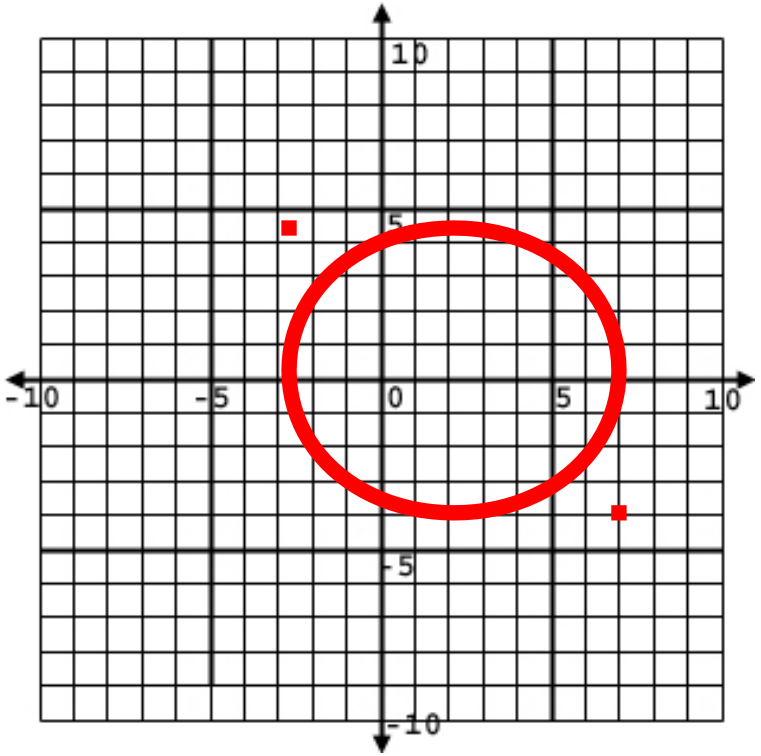
Not a Function



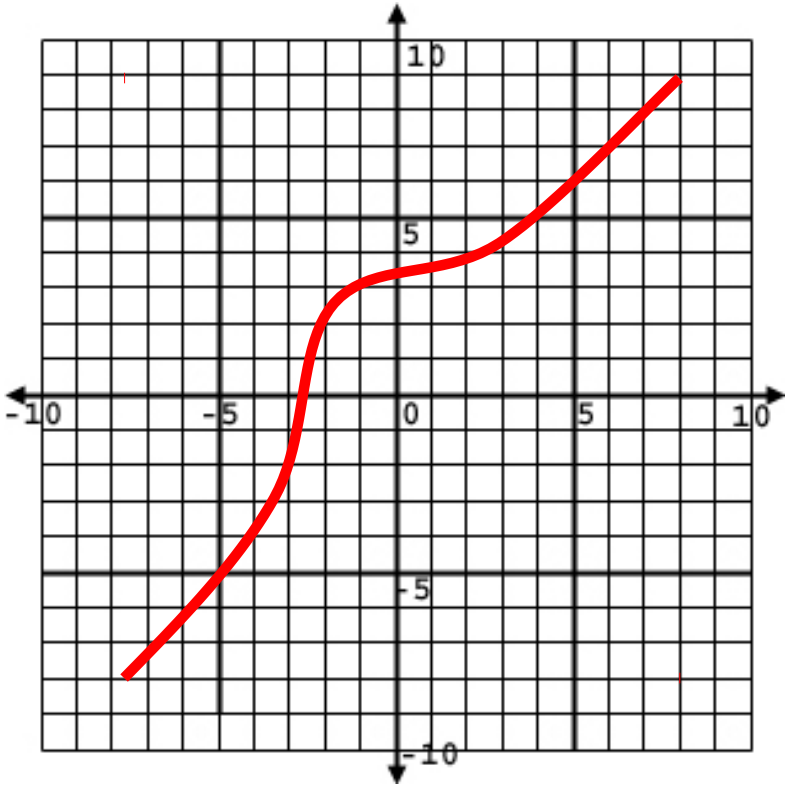
Function



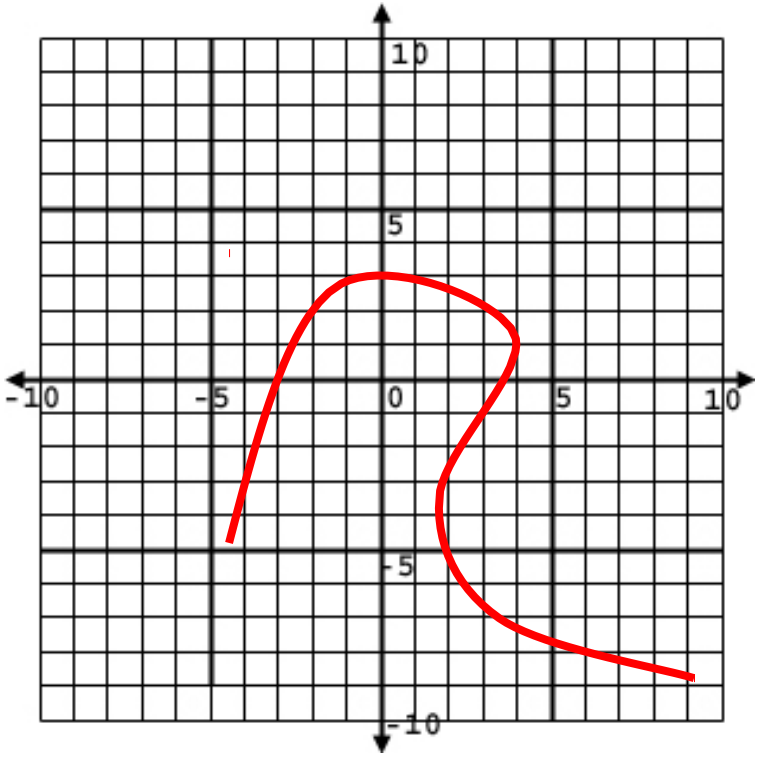
Not a Function



Function



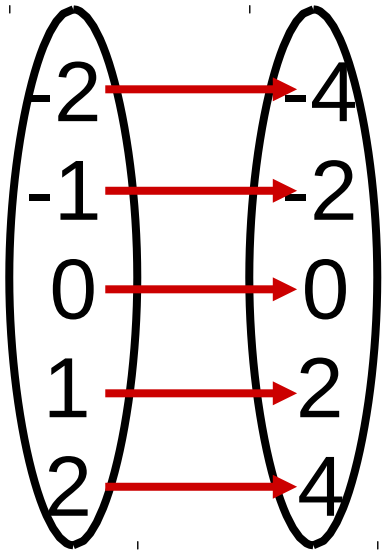
Not a Function



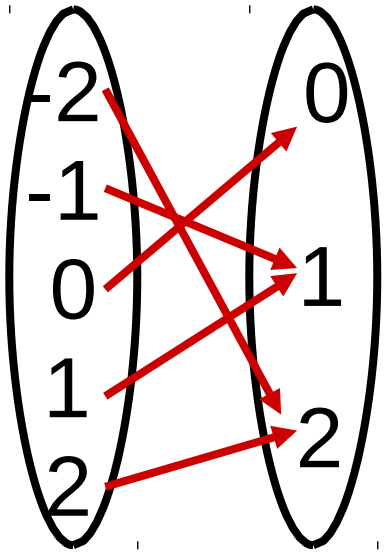
EXAMPLE:

Identify if the given relation is function or not.

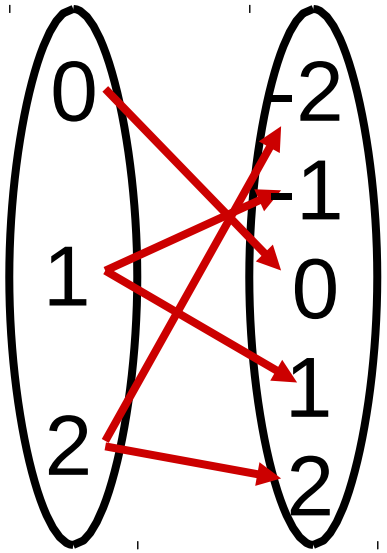
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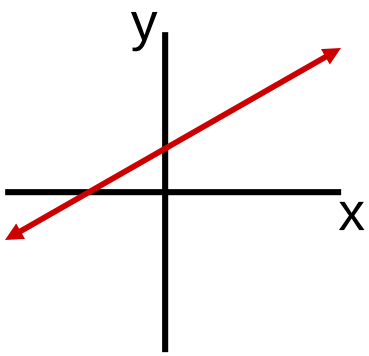
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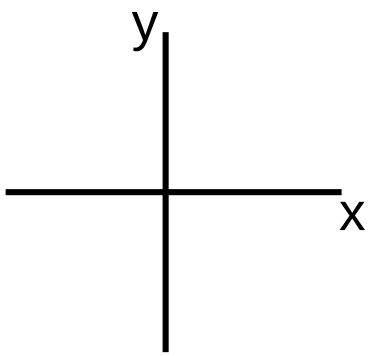


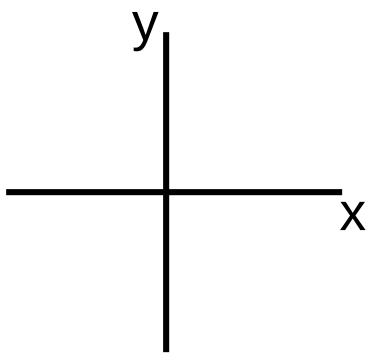
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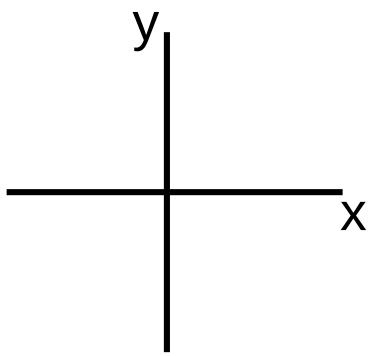


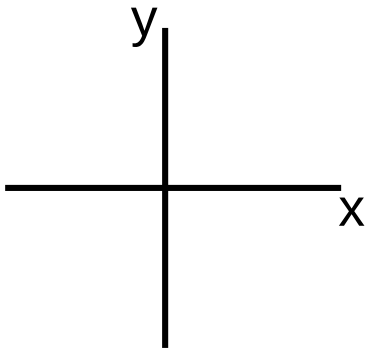
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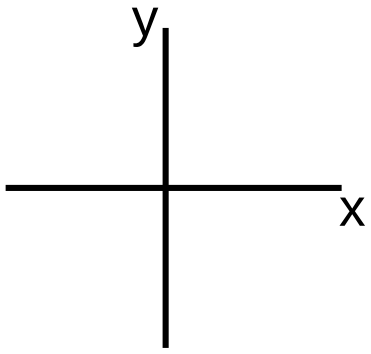


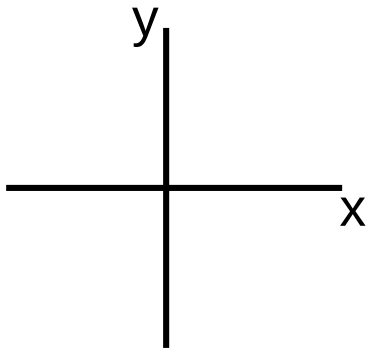


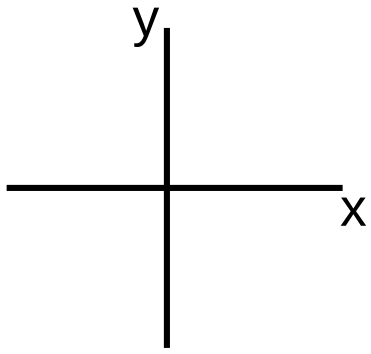




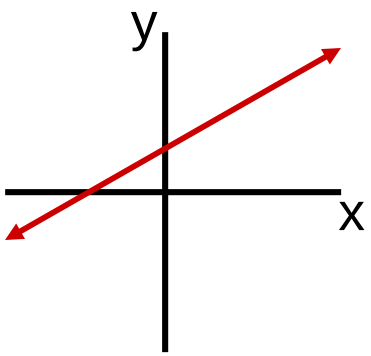




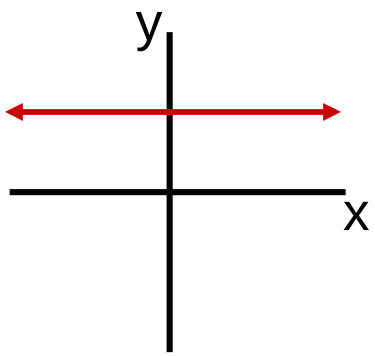


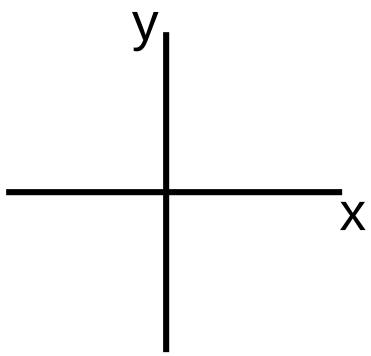


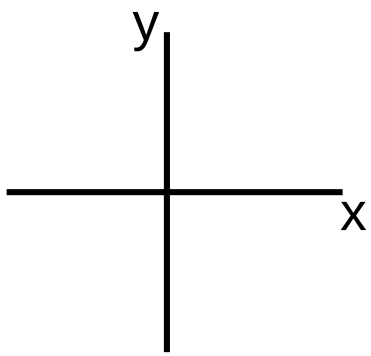
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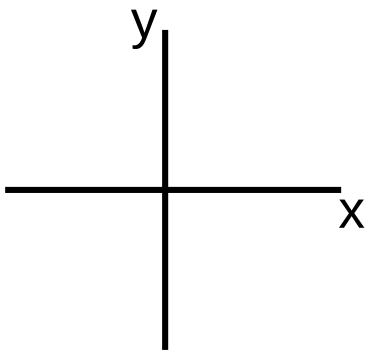


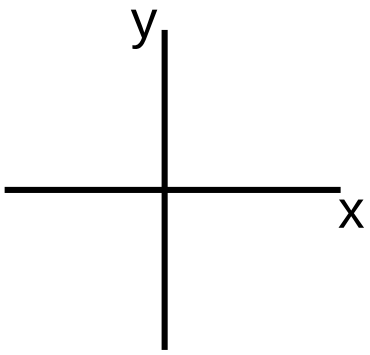
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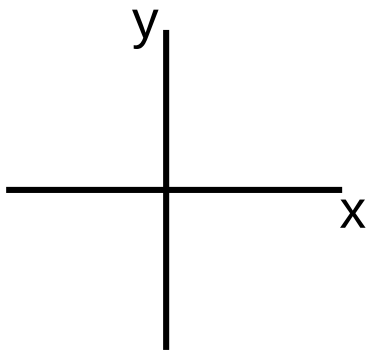


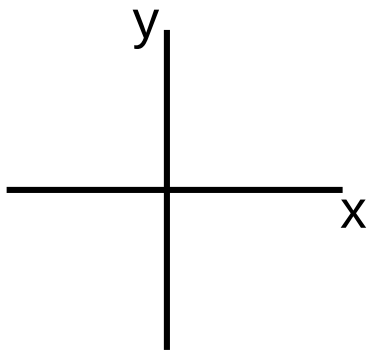




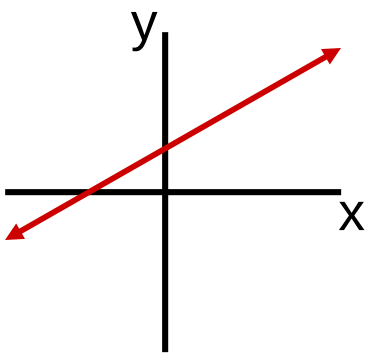




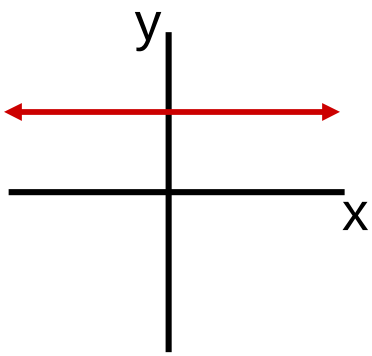




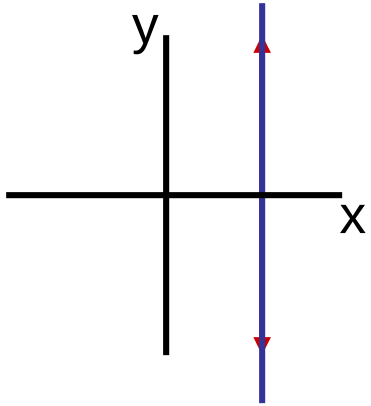
4.

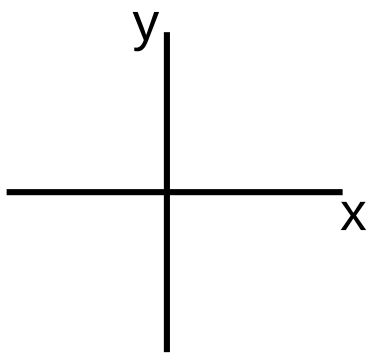


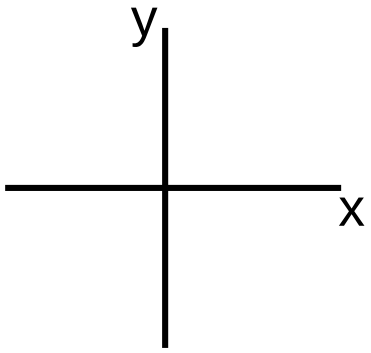
5.

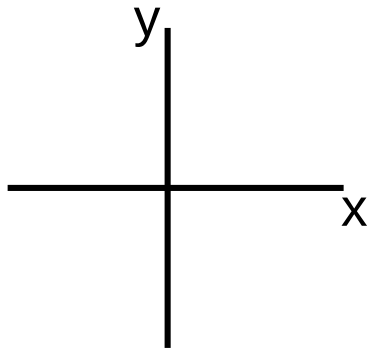


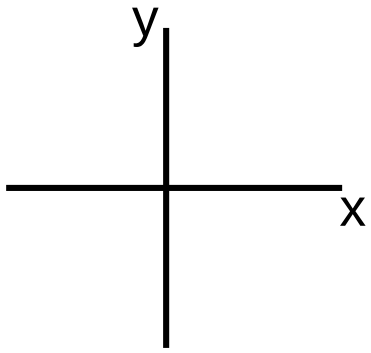
6.

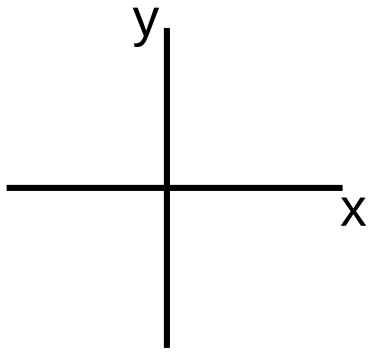




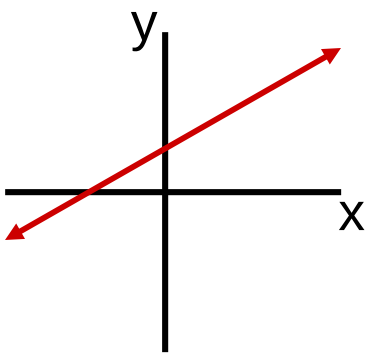




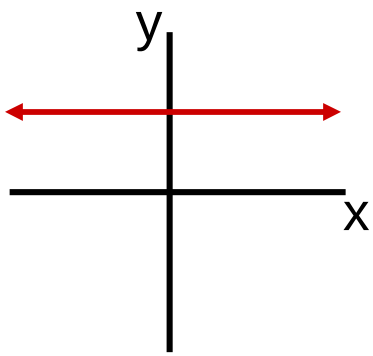




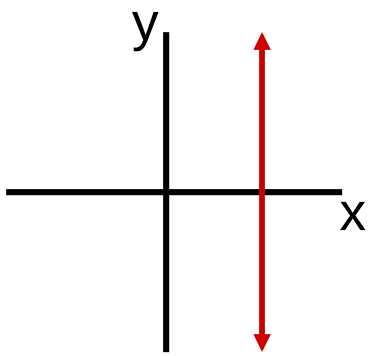
4.



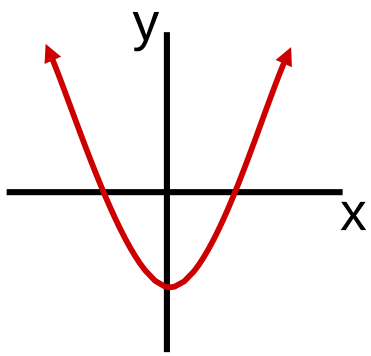
5.

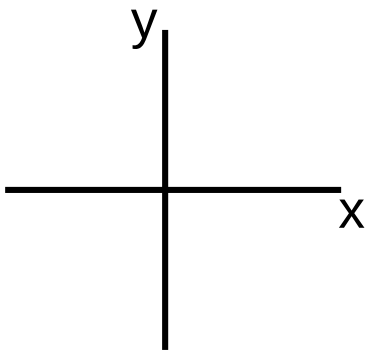


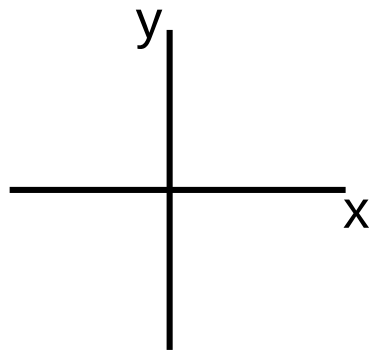
6.

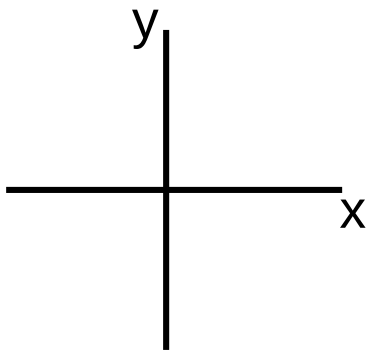


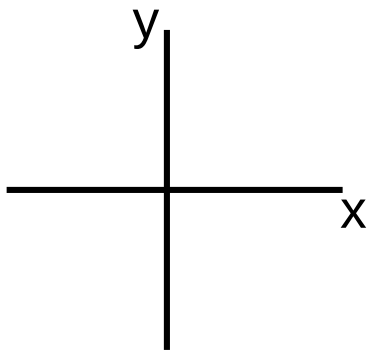
7.



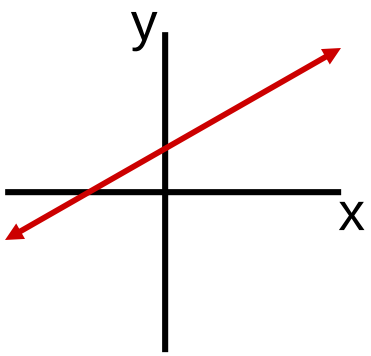




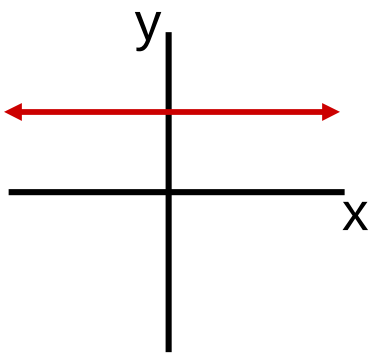




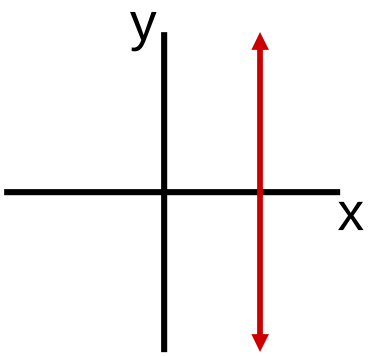
4.



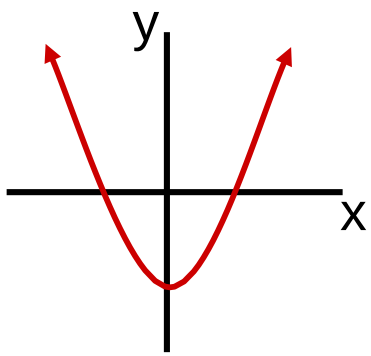
5.



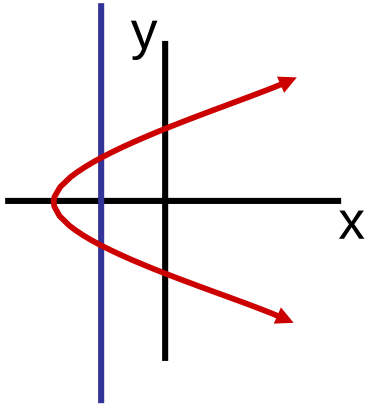
6.



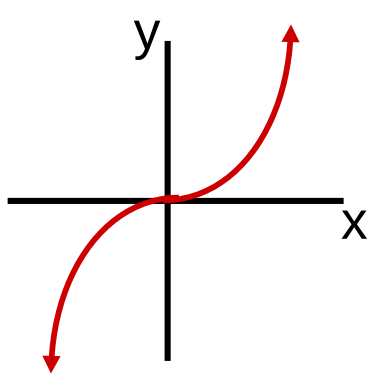
7.



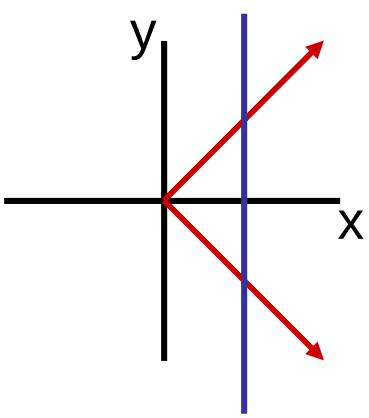
8.



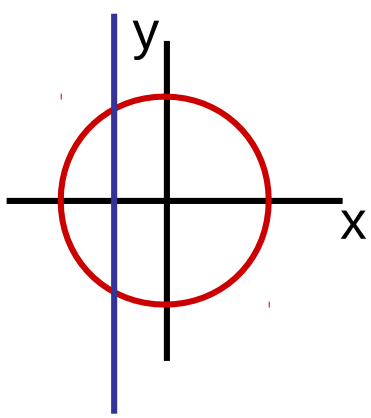
9.



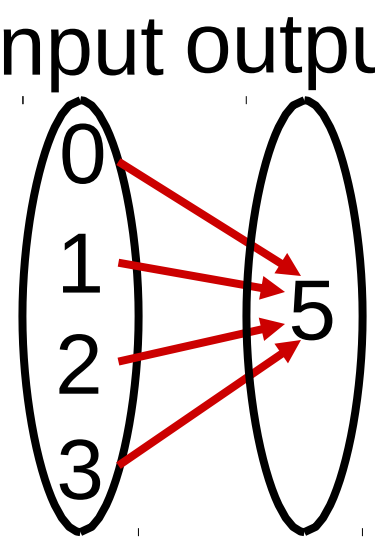
10.



11.



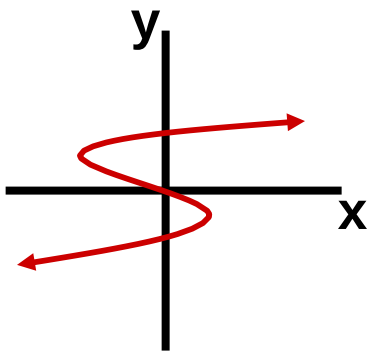
12.



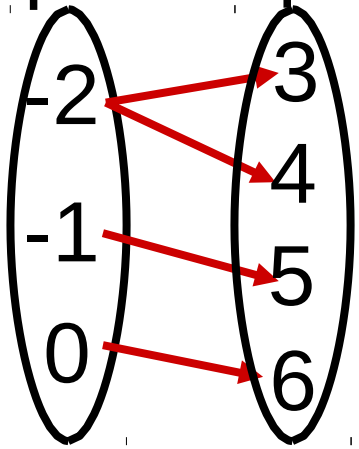
13.

x	y
-3	-1
-3	0
-3	1
-3	2

14.



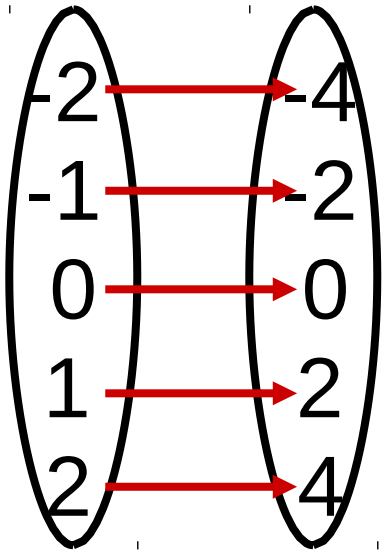
15.



Let’s Check!

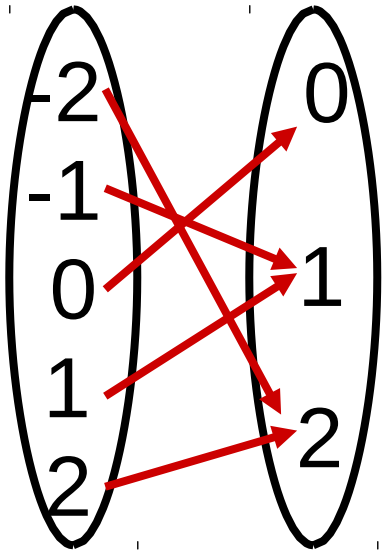
Answers

1.



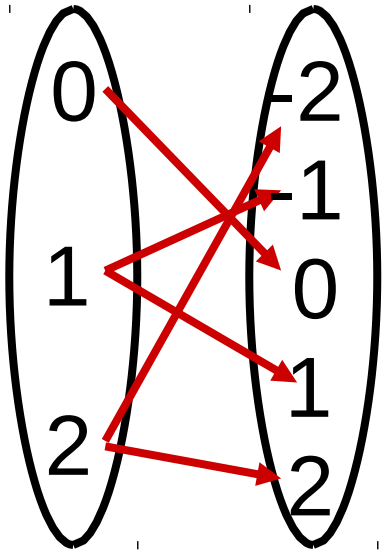
Function

2.



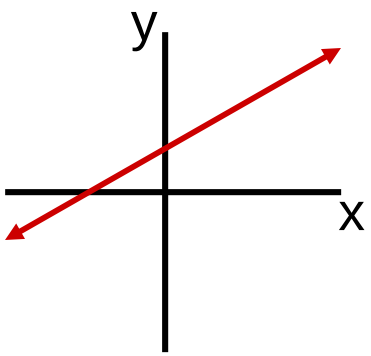
Function

3.



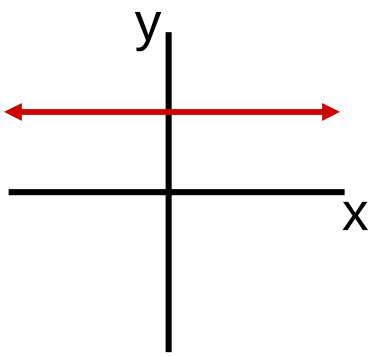
Not a Function

4.



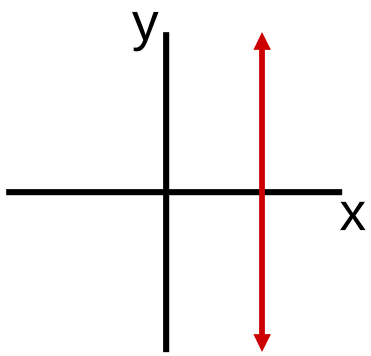
Function

5.



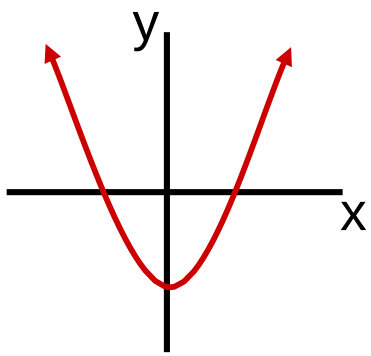
Function

6.



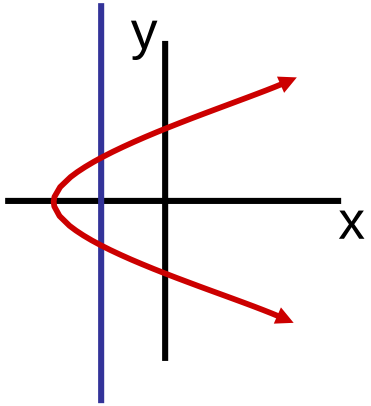
Not a Function

7.



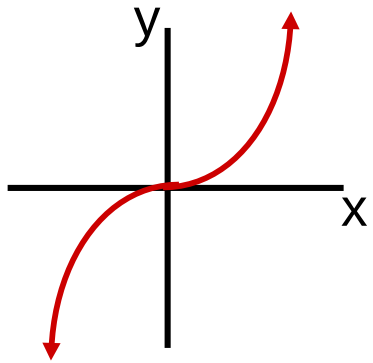
Function

8.



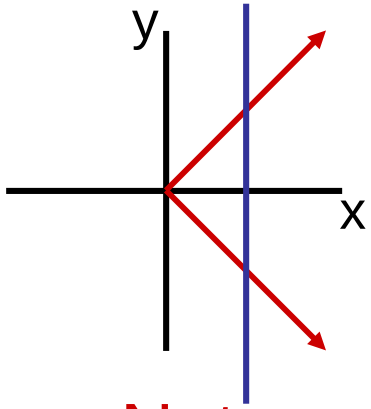
Not a Function

9.



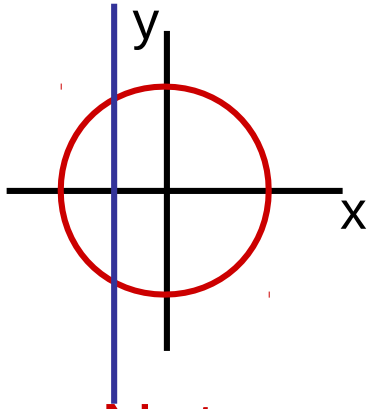
Function

10.



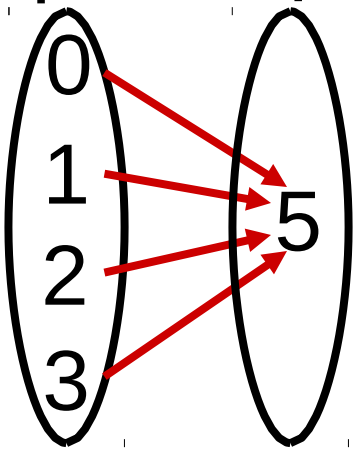
Not a Function

11.



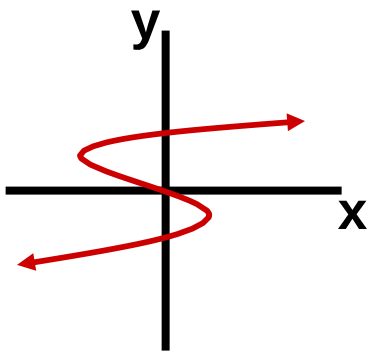
Not a Function

12. input output



Function

14.



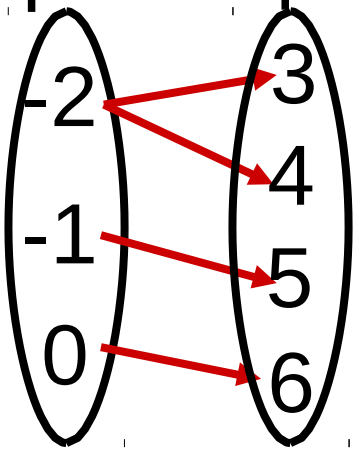
Not a Function

13.

x	y
-3	-1
-3	0
-3	1
-3	2

Not a Function

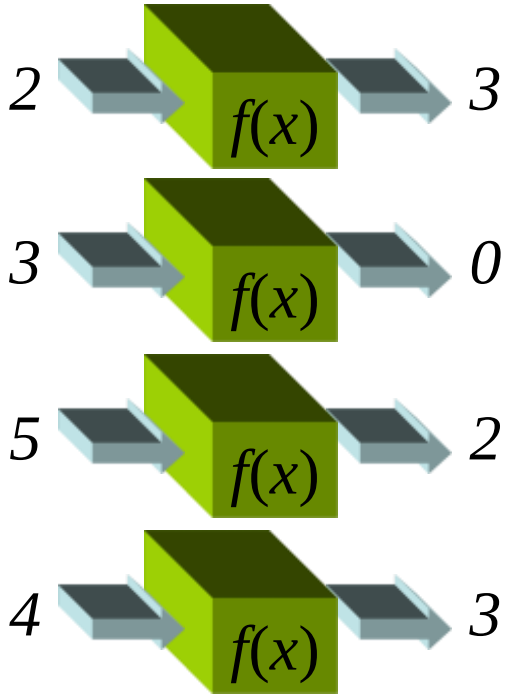
15. input output



Not a Function

Determine whether each relation is a function.

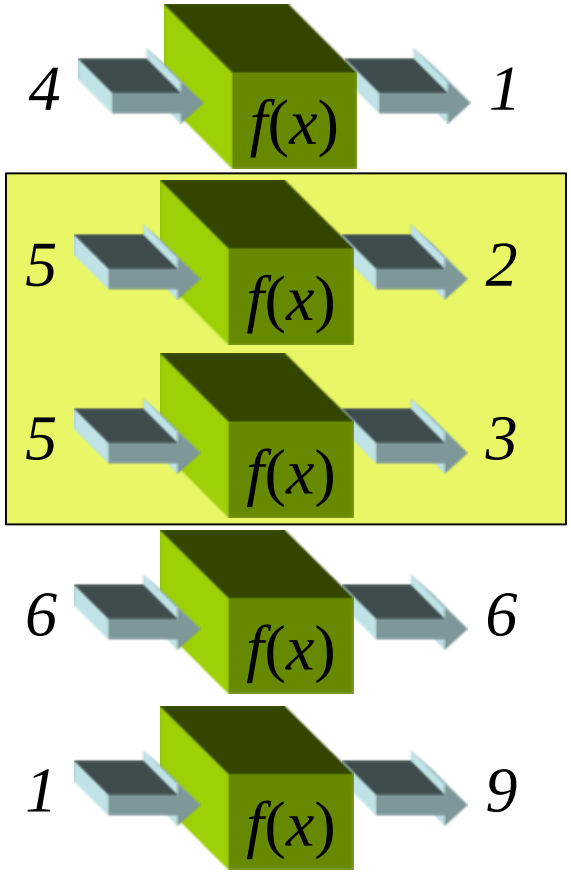
1. {(2, 3), (3, 0), (5, 2), (4, 3)}



YES, every domain is different!

Determine whether the relation is a function.

2. {(4, 1), (5, 2), (5, 3), (6, 6), (1, 9)}



NO,

5 is paired with 2 numbers!

Is this relation a function?

{(1,3), (2,3), (3,3)}

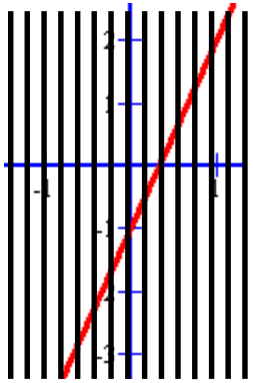
1. Yes
2. No

Answer Now

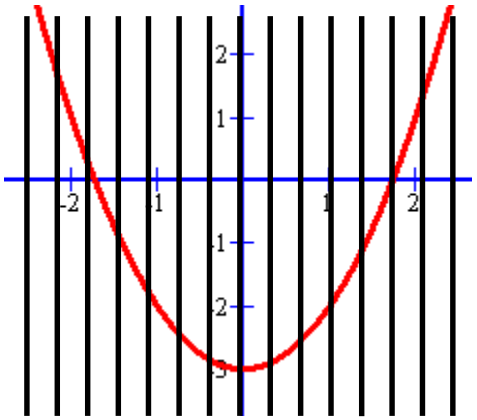
Vertical Line Test (pencil test)

If any vertical line passes through more than one point of the graph, then that relation is not a function.

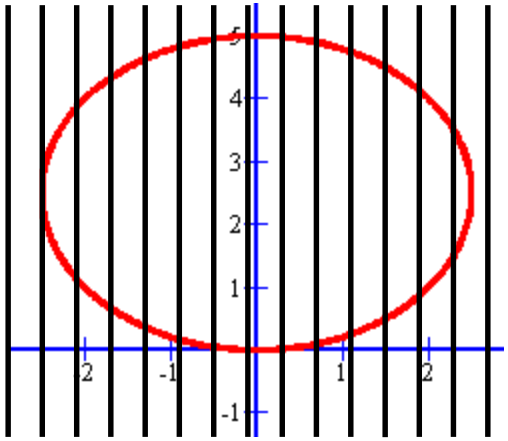
Are these functions?



FUNCTION!

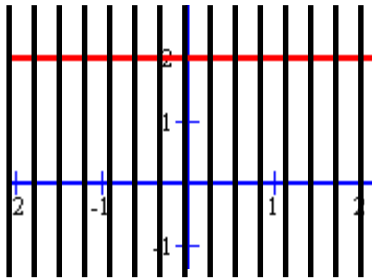


FUNCTION!

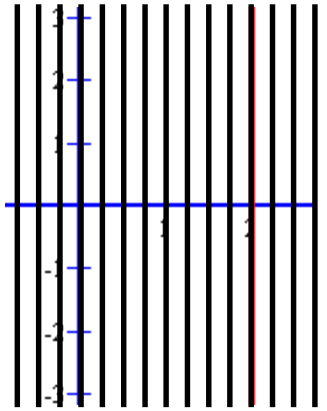


NOPE!

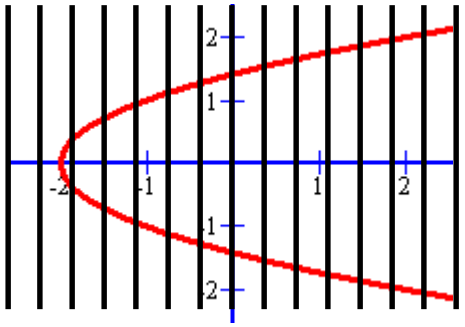
Vertical Line Test



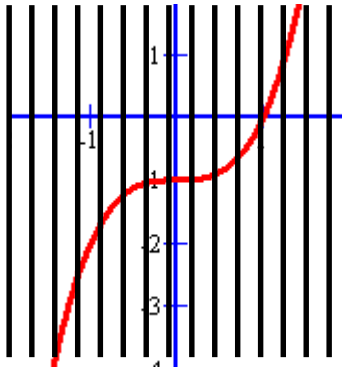
FUNCTION!



NO!

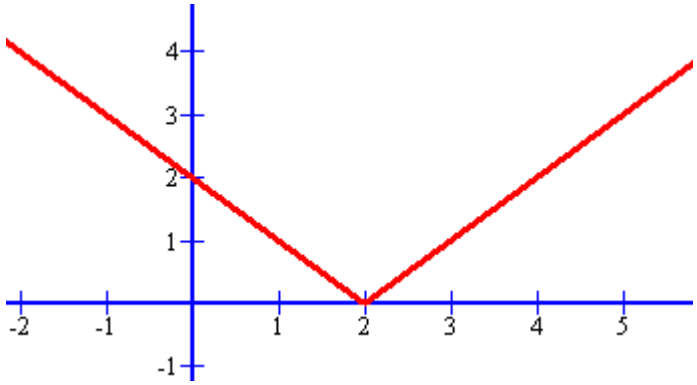


NO!



FUNCTION!

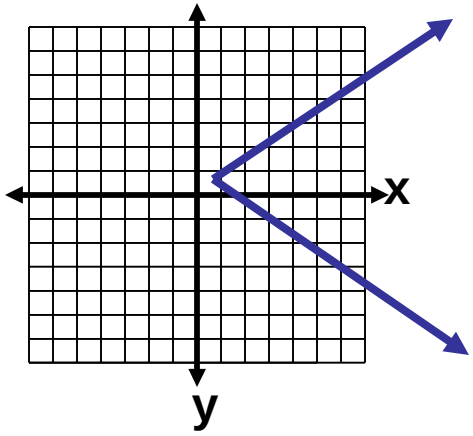
Is this a graph of a function?



1. Yes
2. No

Answer Now

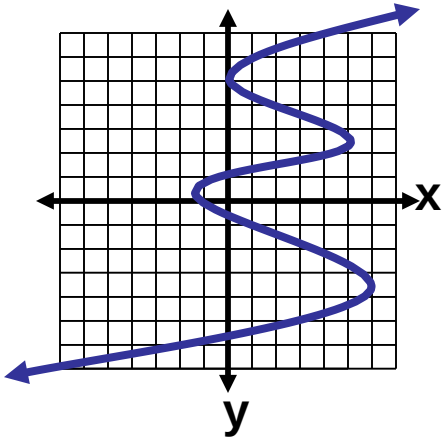
Does the graph represent a function? Name the domain and range.



No

D: $x \geq 1/2$

R: all real numbers

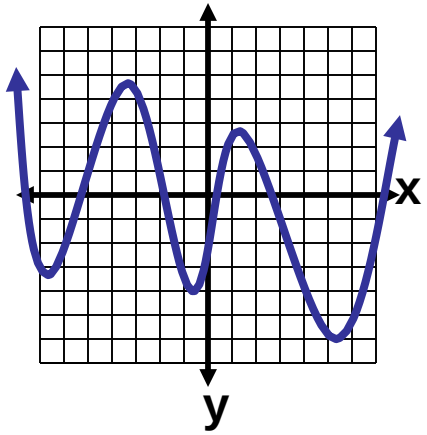


No

D: all real numbers

R: all real numbers

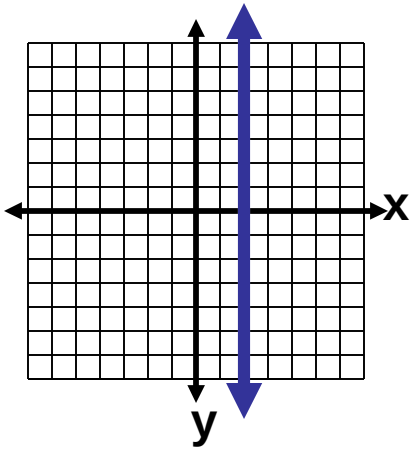
Does the graph represent a function? Name the domain and range.



Yes

D: all real numbers

R: $y \geq -6$



No

D: $x = 2$

R: all real numbers

End of Lesson